Digital Avionics
航電系統

Lecture 5

Prof. Ying-Chih Lai

Textbook

- Reading:
  - Chapter 4: Fly-by-Wire Flight Control
  - Section 4.1~4.5, pp. 179-241

Fly-by-Wire Flight Control

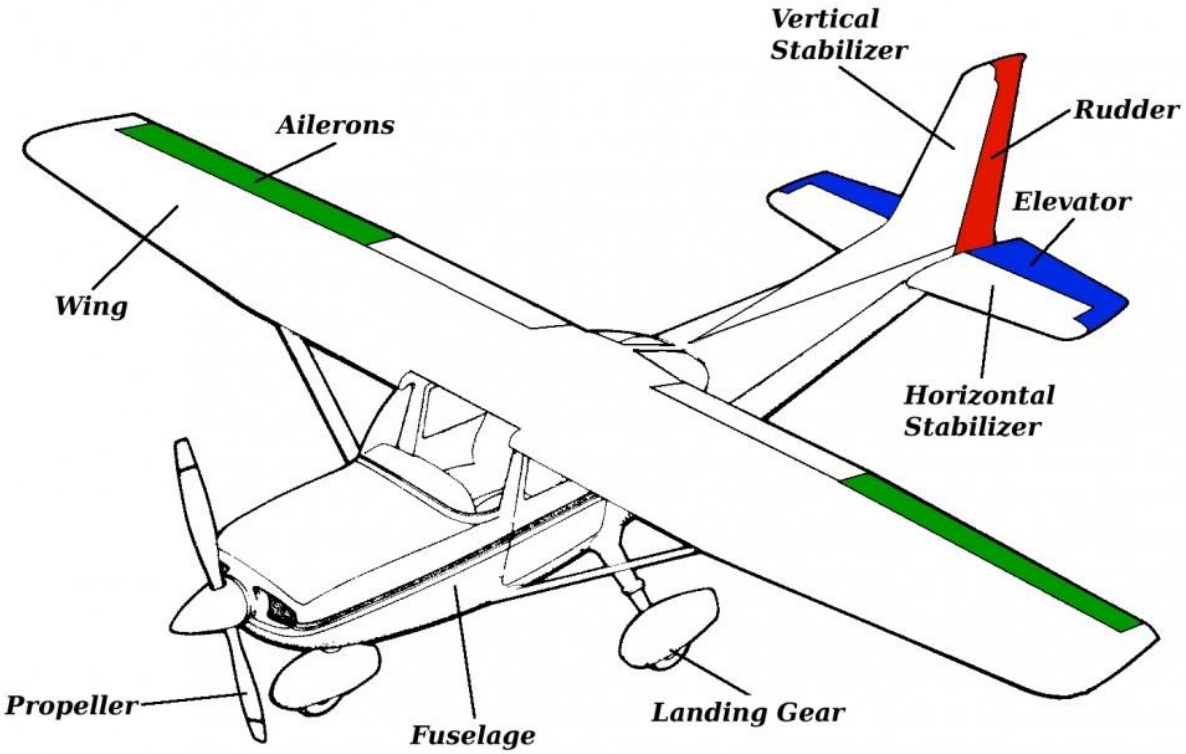
Fly-by-Wire (FBW) Flight Control

- The introduction of fly-by-wire (FBW) flight control systems has been a watershed development in aircraft evolution as it has enabled technical advances to be made which were not possible before.
- One of the unique benefits of a FBW system is the ability to exploit aircraft configurations which provide increased aerodynamic efficiency, like more lift and lower drag, but at a cost of reduced natural stability.
- The FBW system provides high-integrity automatic stabilization of the aircraft to compensate for the loss of natural stability and thus enables a lighter aircraft with a better overall performance to be produced compared with a conventional design.

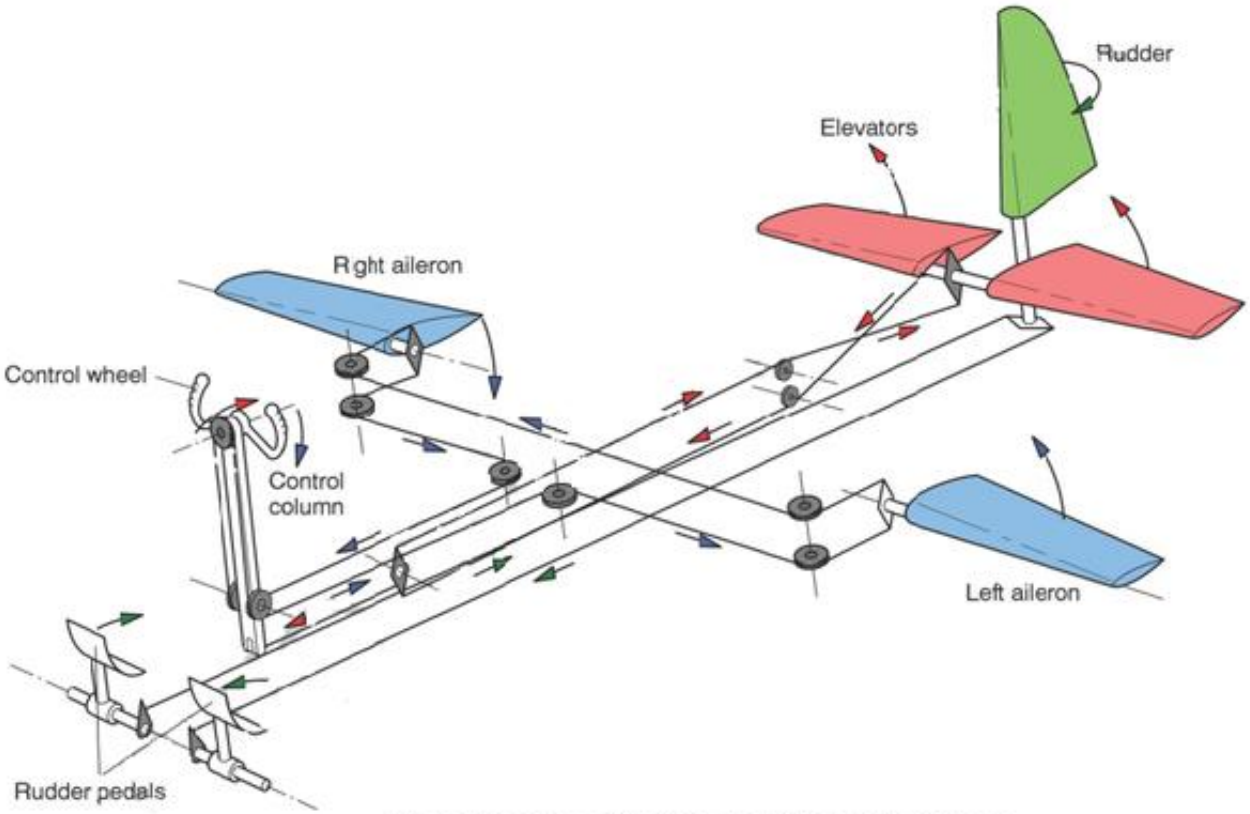
Fly-by-Wire (FBW) Flight Control

- Aircraft with FBW flight control systems first came into service in the late 1970s using analog implementation. Digital FBW systems have been in service since the late 1980s.
- A major factor has been the development of failure survival digital flight control systems and their implementation in VLSI microcircuits (Semiconductor).
- The current generation of civil airliners exploit FBW control, such as Airbus A320, A340, A380, and Boeing 777 and 787.

Control Mechanic Design



Control Mechanic Design



Movements of Primary Flight Controls

Nature of FBW

- Aircraft is usually designed with naturally stable system structure. As long as speed and attitude can be maintained, it can return to stable condition automatically.
- But for higher performance in military aircraft to escape from attacks, aircraft sacrifices naturally stable characteristics for better operation and performance. It becomes unstable in nature.

Performance to Stability

- Military Aircraft sacrifices natural stability to receive high performance.
- Wing design changes this nature and becomes deterministic to overall system characteristics.
- FBW can overcome the defect and help pilot to maneuver aircraft easily (reducing workload and increasing flight safety).

Unstable Aircraft









Unstable Aircraft







ABOVE A Westland windtunnel model of Geoffrey Hill's proposed 63ft-span Pterodactyl Mk VII flying-boat, powered by four 120 h.p. Gipsy III engines, arranged in a "push-me-pull-you" configuration.



Requirements of FBW

- FBW system is designed to solve the unstable system problem for high performance aircraft applying control technologies.
- Use high speed computer to calculate feedback signals, command terminal actuators via electronic signals for precise and fast decisions to ensure system stability
- FBW indeed uses electronic signals to replace mechanical levers for fast response.

Basic Elements of a FBW Flight Control

- Eurofighter Typhoon

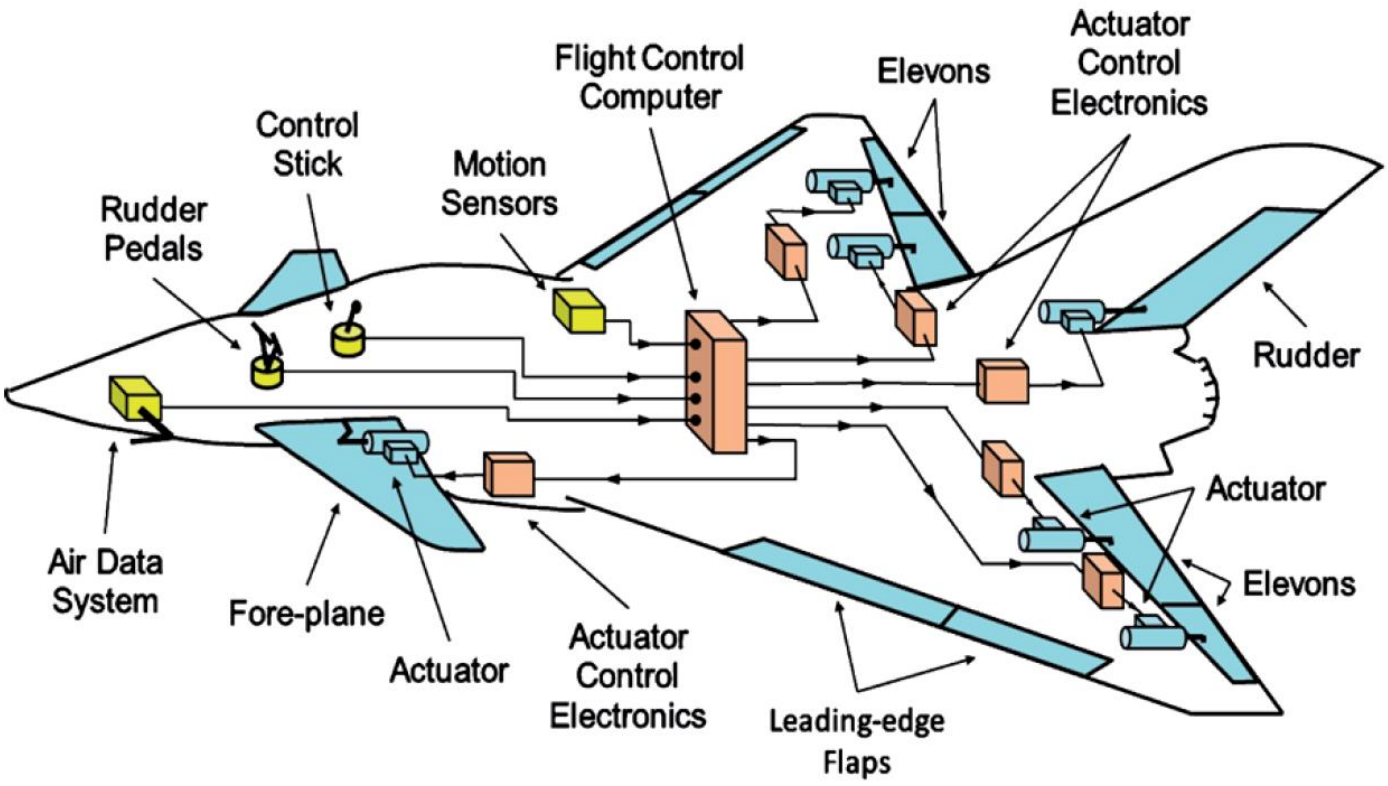


Fig. 4.1 Basic elements of a FBW flight control system.

Operation of FBW

- Modern aircraft uses FBW to reduce pilot load in flight control
- FBW directly combines computer with hardware and software to an integral operation



NASA's F-8C digital fly-by-wire testbed

Enhancement of FBW

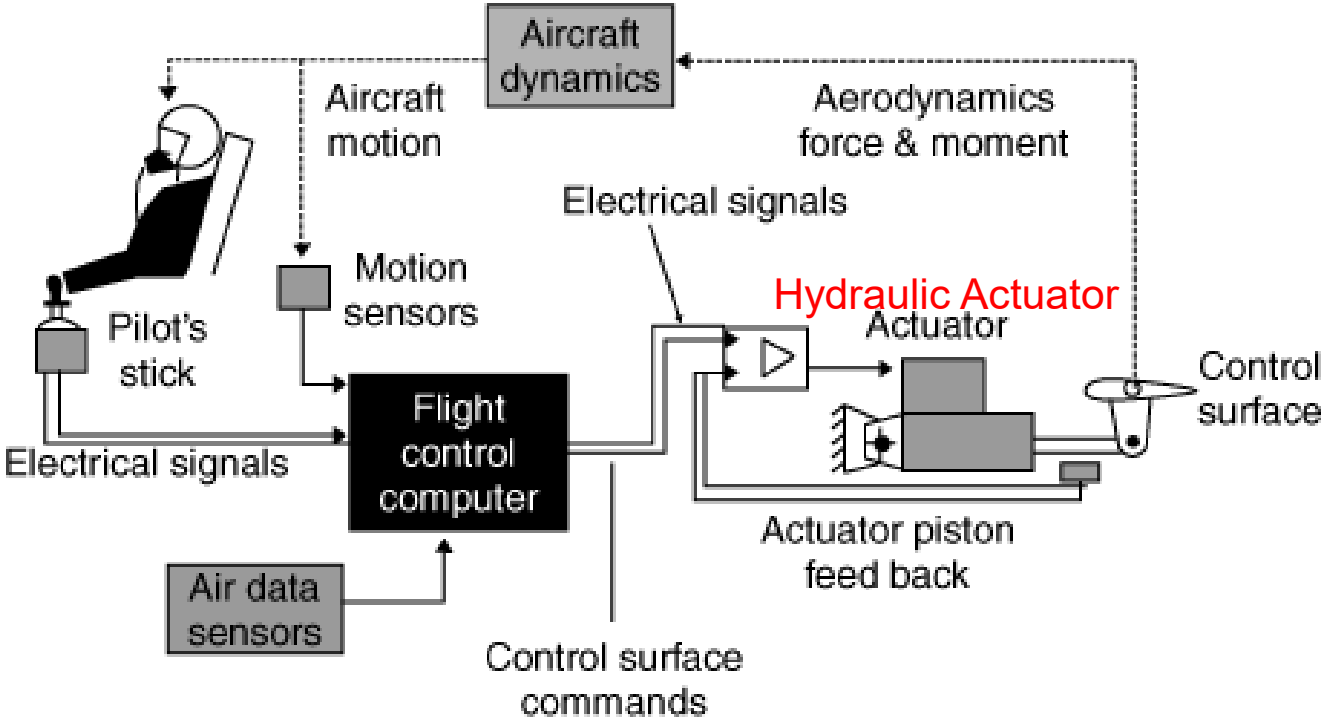
- (1) Enhance the capability to monitor aerodynamic effects and aircraft response
- (2) Need less energy to operate aircraft from effective navigation aids
- (3) Reduce aircraft weight
- (4) Simplify operation procedure, elevate operation efficiency in flight performance of Pitch, Roll, Yaw, and throttle
- (5) Ease of Integration of the Autopilot

Effect of FBW

- (1) Signal transmits faster and better to activate actuators more efficiently
- (2) Shorten time delay in control command with more precise feedback control
- (3) Transparent in control and command process
- (4) Effective redundant system to avoid fault or failure
- (5) Install hierarchical computer structure to share analysis and record

Structure of FBW

- Pilot commands all terminal elements via electronic signals, no actual force is required



Basic Elements of a FBW Flight Control

- The total elimination of all the complex mechanical control runs and linkages
- The interposition of a computer between the pilot's commands and the control surface actuators.
- The aircraft motion sensors which feed back the components of the aircraft's angular and linear motion to the computer.
- The air data sensors which supply height and airspeed information to the computer.
- Not shown in the figure is the redundancy to enable failures in the system to be absorbed. The flaps can also be continually controlled by the Flight Control Computer.

Electrical Data Transmission

- Electrical transmission of signals and commands is a key element in a FBW system. Modern systems use a serial digital data transmission system with time division multiplexing.
- Figure 4.2 shows how a digital flight control system is interconnected using a digital data bus.
- Military FBW generally use the well established Mil Std 1553 databus system.
- The Boeing 777 uses the ARINC 629 data bus system.

Flight Control System Bus Configuration

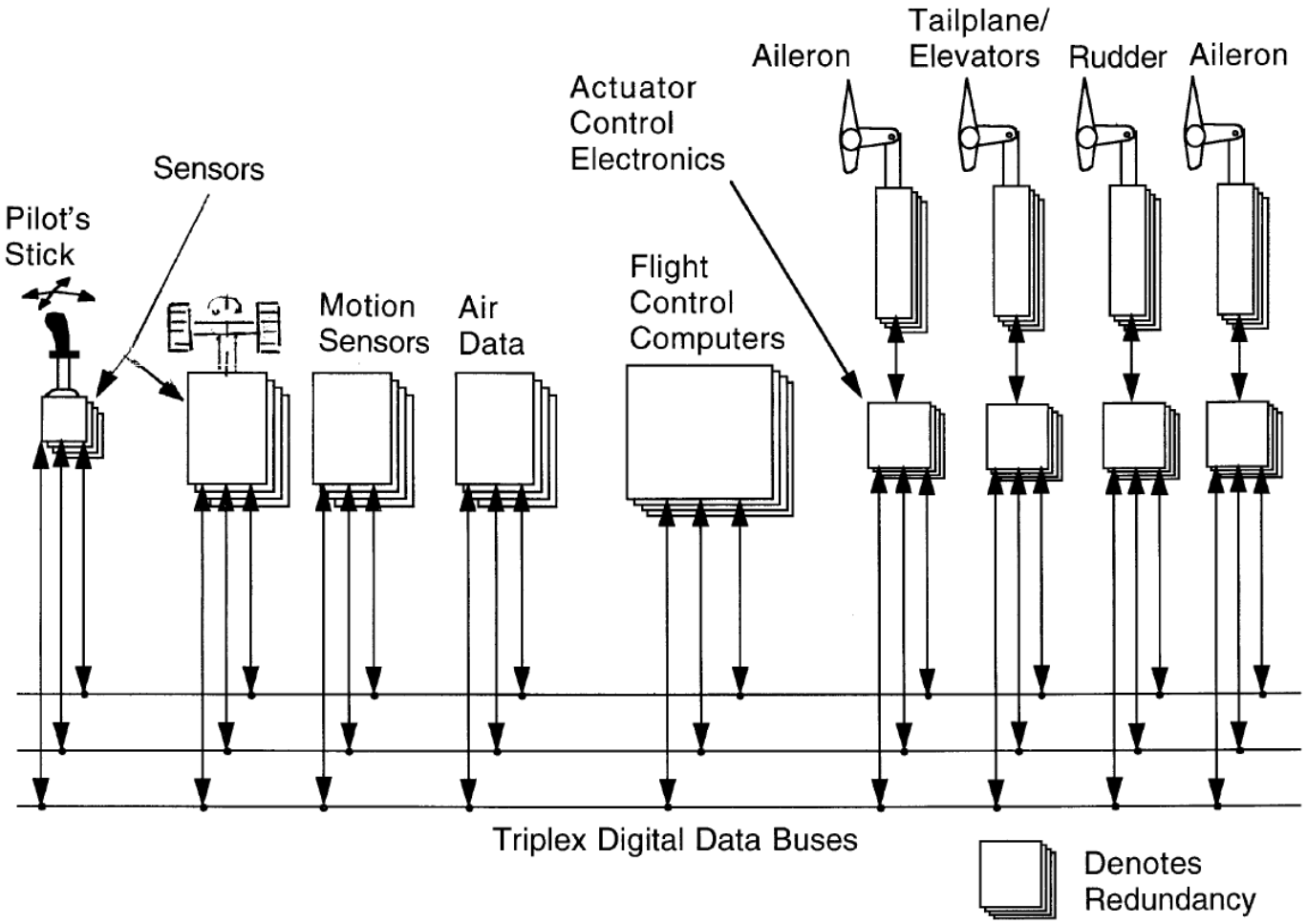


Fig. 4.2 Flight control system bus configuration.

Motion Sensing in FBW

- Rate gyros which measure the angular rates of rotation of the aircraft about its pitch, roll and yaw axes.
- Linear accelerometers which measure the components of the aircraft's acceleration along these axes.
- Air data system outputs altitude and indicated air speed and etc.
- Air flow sensor measures longitudinal and lateral wing control angle

FBW Flight Control System

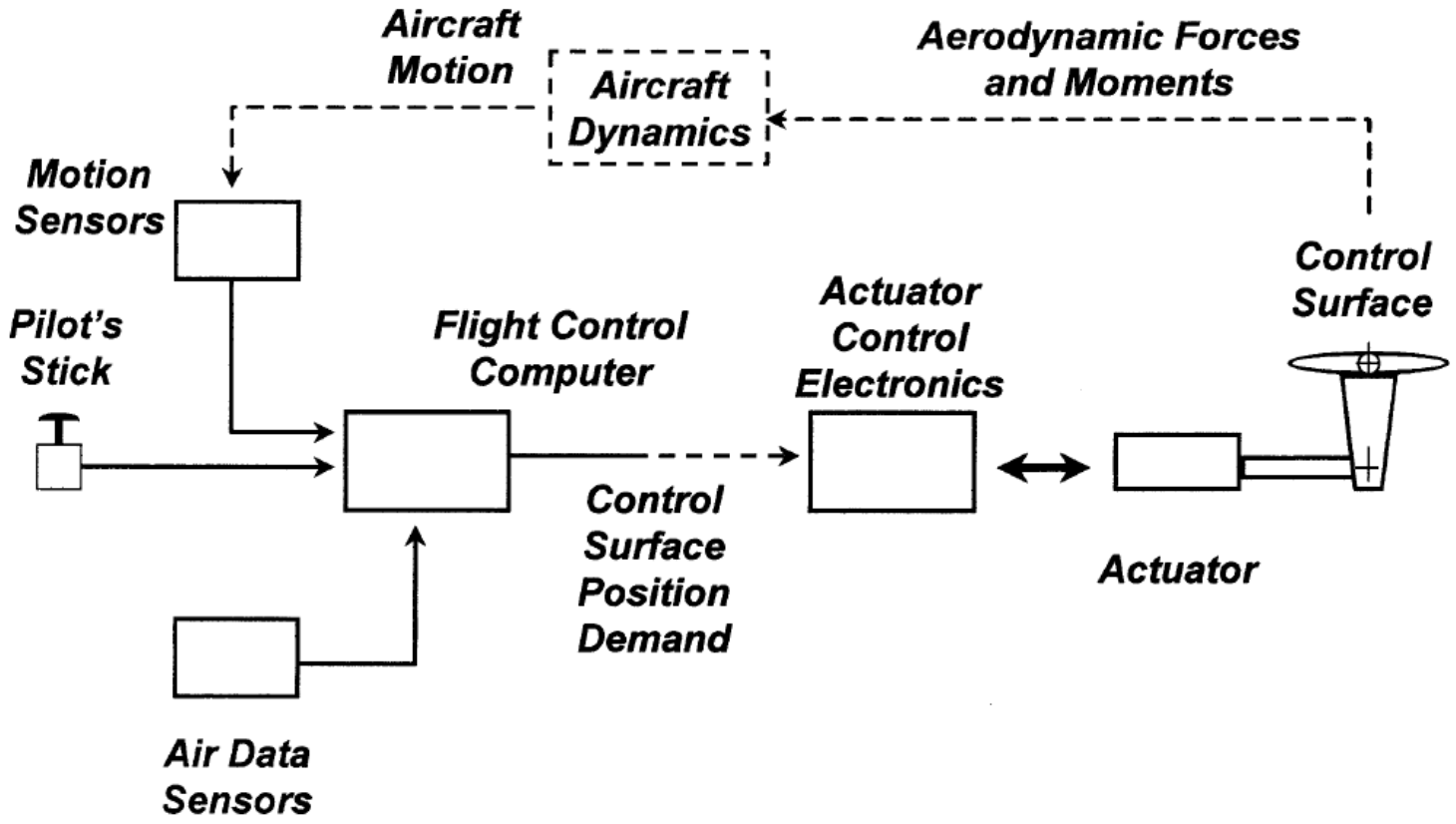


Fig. 4.4 Fly by wire flight control system.

Inputs for FBW

- FBW computer inputs signal sources as:
  - Control stick or lever from pilot in electronic signal
  - Sensor measurements for aircraft dynamic signals
  - Air data system output of 7 data as altitude, speed, climb rate, Mach, temperature, etc.

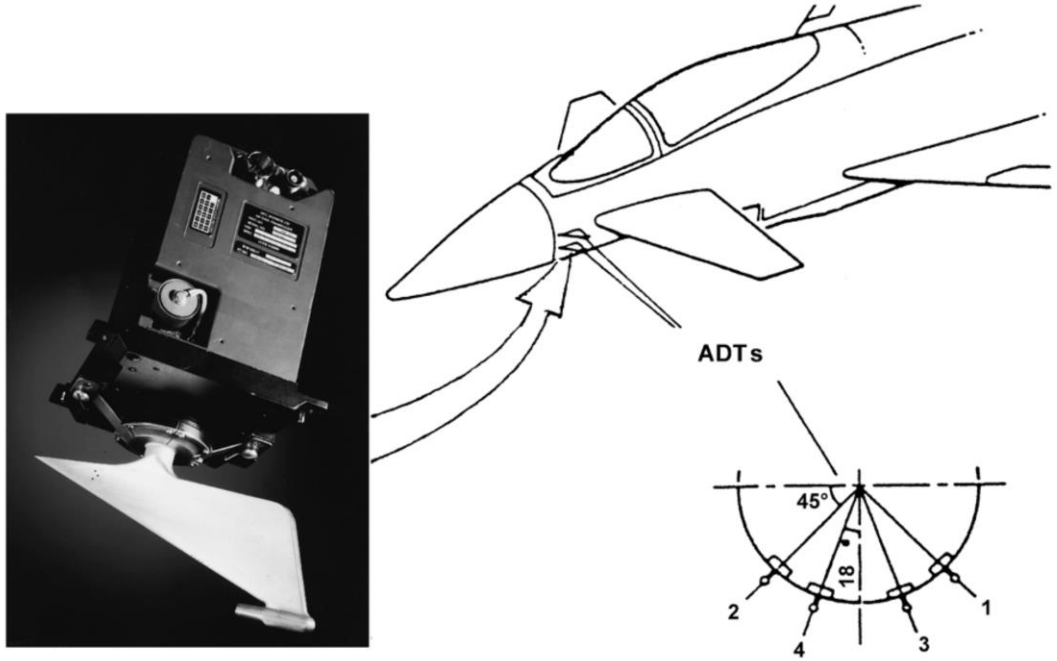


Fig. 4.5 Integrated air data transducer system (courtesy of BAE Systems).

Automatic Control Loop in FBW

- Automatic feedback control in FBW improves flight performance as:
  - The input/output has no direct cause/effect relationship to system gain
  - System bandwidth can be controlled by delay controller
  - Appropriate damping effect is added on

Controller of FBW

- Input from control stick by pilots
- Control stick is the same as joy stick in games, but add on force buffer to avoid overwhelming
- Additional spring is attached for virtual force to let pilot “feel” the control stick motion

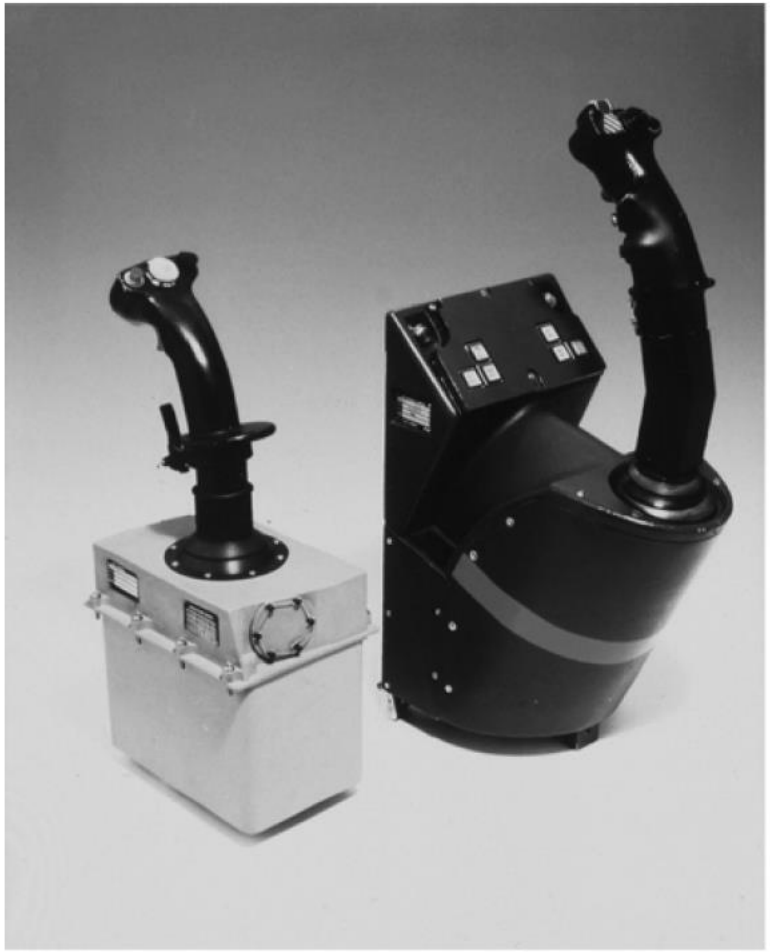
Flight Control Computer

- The tasks carried out by the computing system comprise:
  - (a) Failure detection.
  - (b) Fault isolation and system reconfiguration in the event of a failure.
  - (c) Computation of the required control surface angles.
  - (d) Monitor aircraft flying in safety region
  - (e) Automatic Built-in Test (BIT) and record

FBW Control Sticks/Passive FBW Inceptors

- A FBW inceptor is defined as a device which translates the pilot's control inputs into electrical signals. They can be divided into two basic types – passive and active.
- The passive type of FBW inceptor provides a fixed stick force–stick displacement relationship by means of a mechanical spring-box arrangement.
- The active type of inceptor, in contrast, can provide a wide range of stick force–stick displacement characteristics by computer control of force motors which back drive the control stick.

Control Stick of FBW



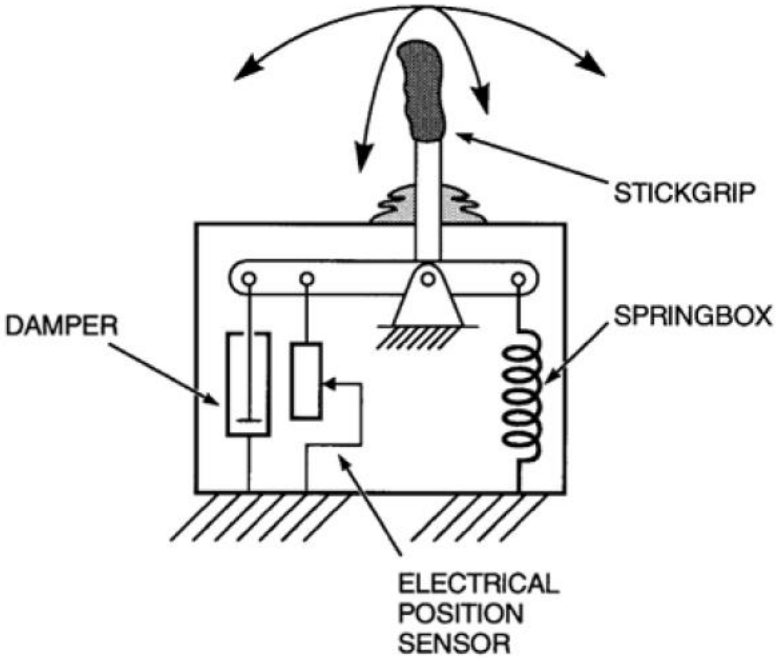
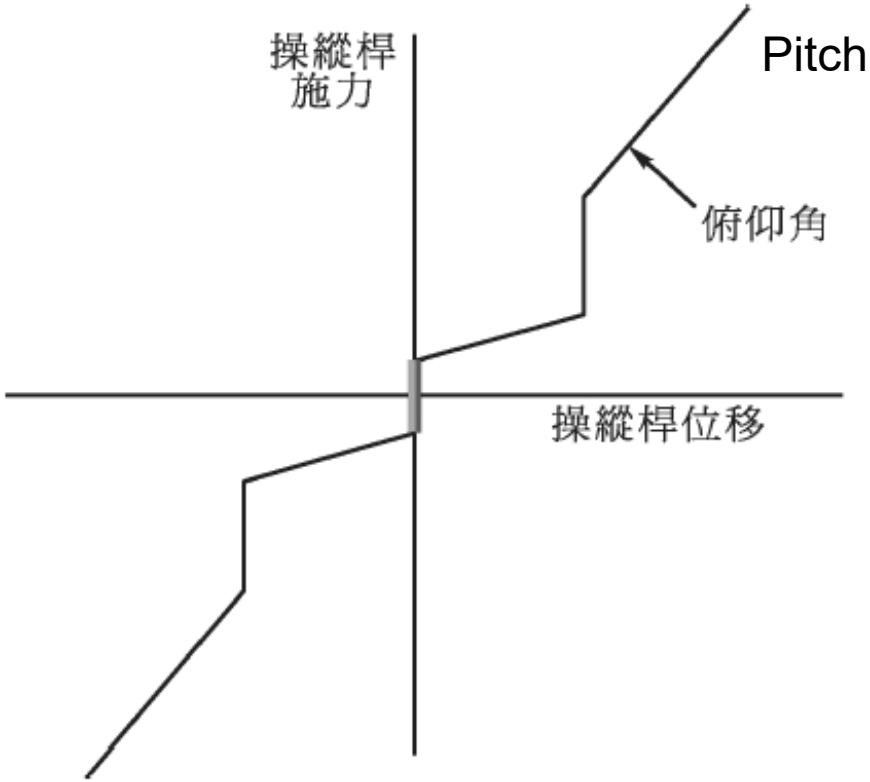
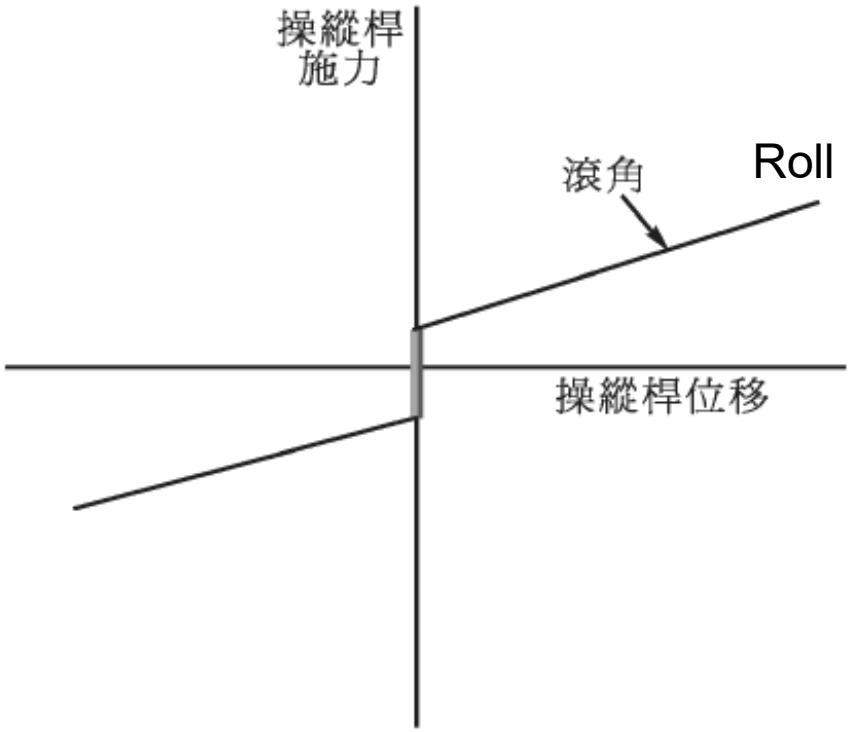


Fig. 4.6 FBW control stick/passive FBW inceptor features.

Roll Control Response

- Control responses from control stick
- These characteristics are carefully tailored to meet the consensus of pilot approval.





Closed-Loop Control in FBW

- A well designed closed-loop control system offers the following advantages over an open-loop control system:
  - (i) The steady-state output to input relationship is substantially independent of changes in the loop gain provided this remains sufficiently high.
  - (ii) The system bandwidth is improved and the phase lag when following a dynamically varying input is reduced.
  - (iii) A fast well damped response, which is little affected by normal changes in the loop gain, can generally be achieved by suitable design of the control loop.

Closed-Loop Control in FBW

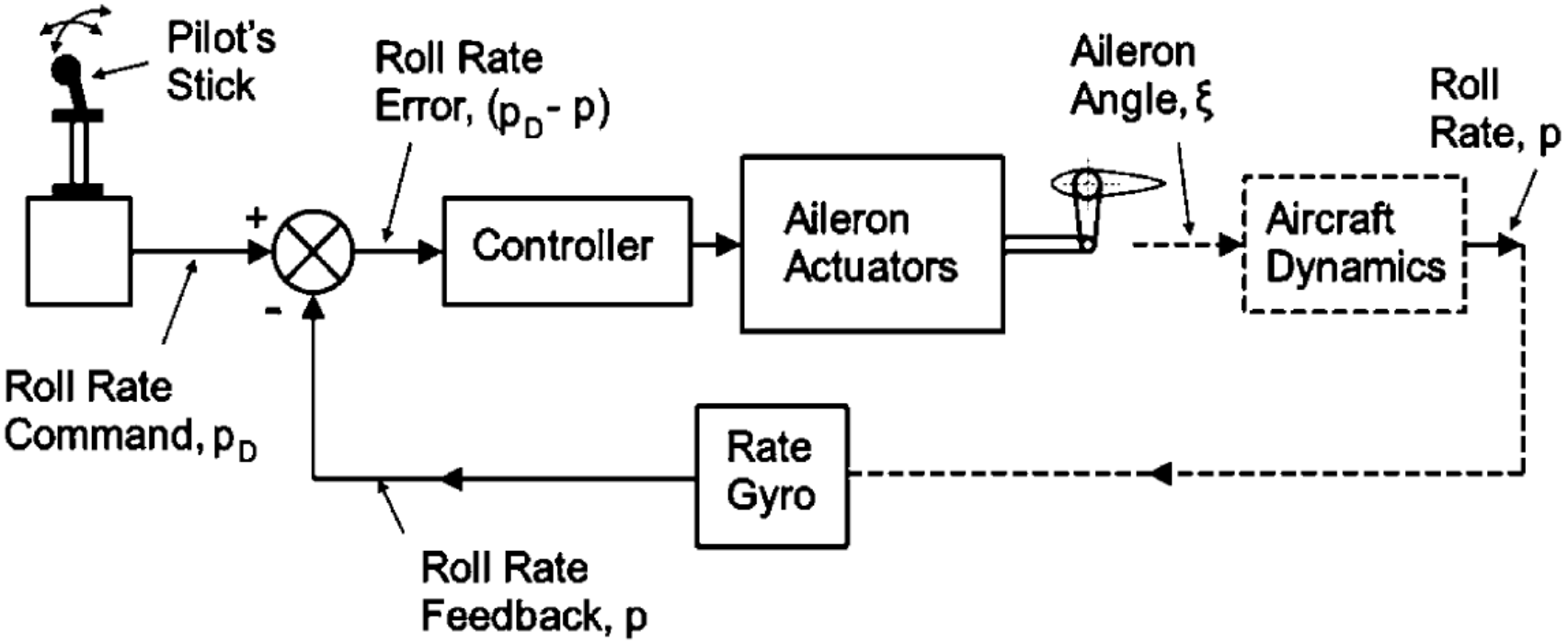
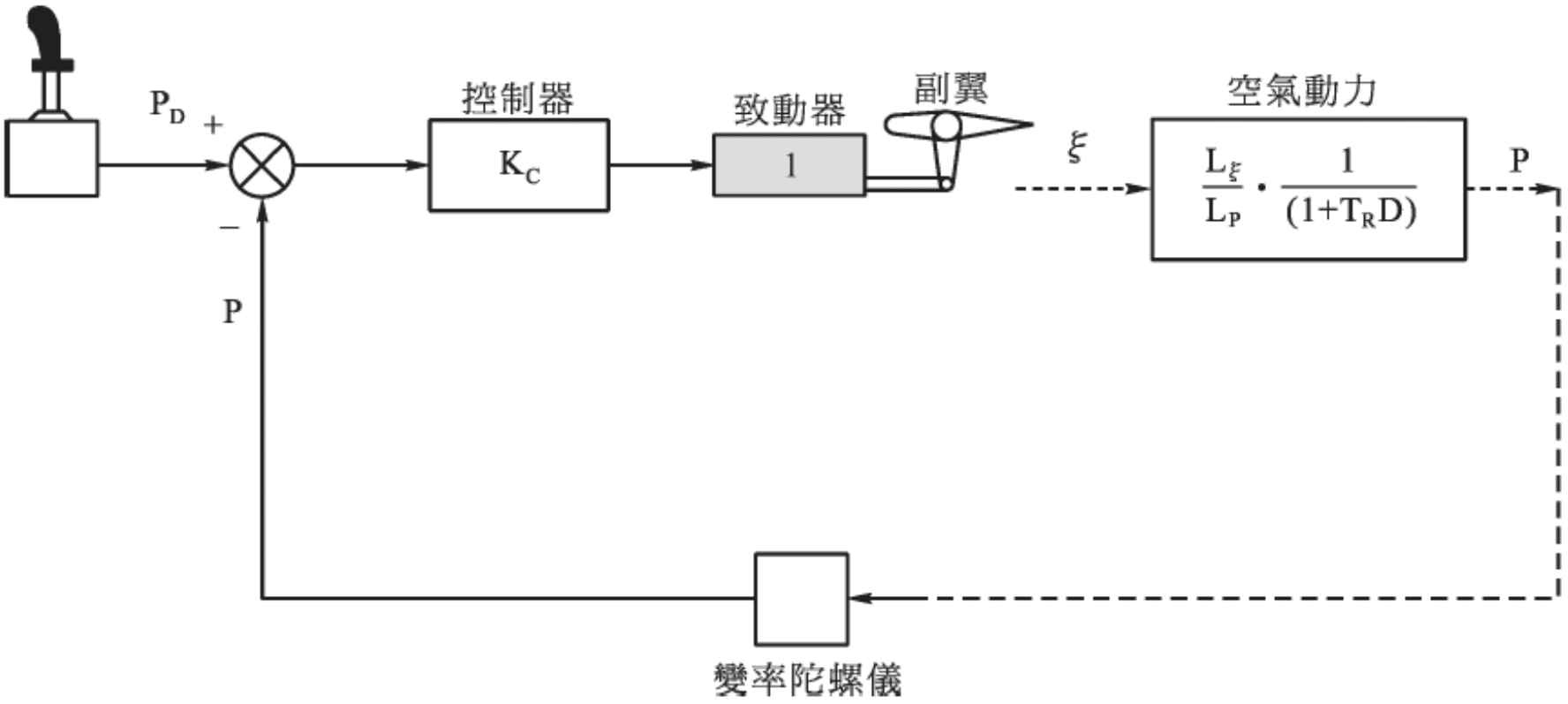


Fig. 4.7 Roll rate command system.

Closed-Loop in Roll Control



Closed-Loop in Roll Control

- The open-loop transfer function is

$$\frac{P}{\xi} = \frac{L_\xi}{L_P}\frac{1}{1+T_R D}$$

- The closed-loop response is given by

$$\frac{P}{P_D} = \frac{\dfrac{K_C L_\xi / L_P}{1+T_R D}}{1+\dfrac{K_C L_\xi / L_P}{1+T_R D}} = \frac{K}{1+K}\bullet\frac{1}{1+\dfrac{T_R}{1+K}D}$$

Closed-Loop in Roll Control

- Aircraft characteristic parameters are changing with altitude, pressure and speed, when K>1, it becomes

$$\frac{P}{P_D} \approx \frac{1}{1+\frac{T_R D}{K}} \approx 1$$

Closed-Loop Control in FBW

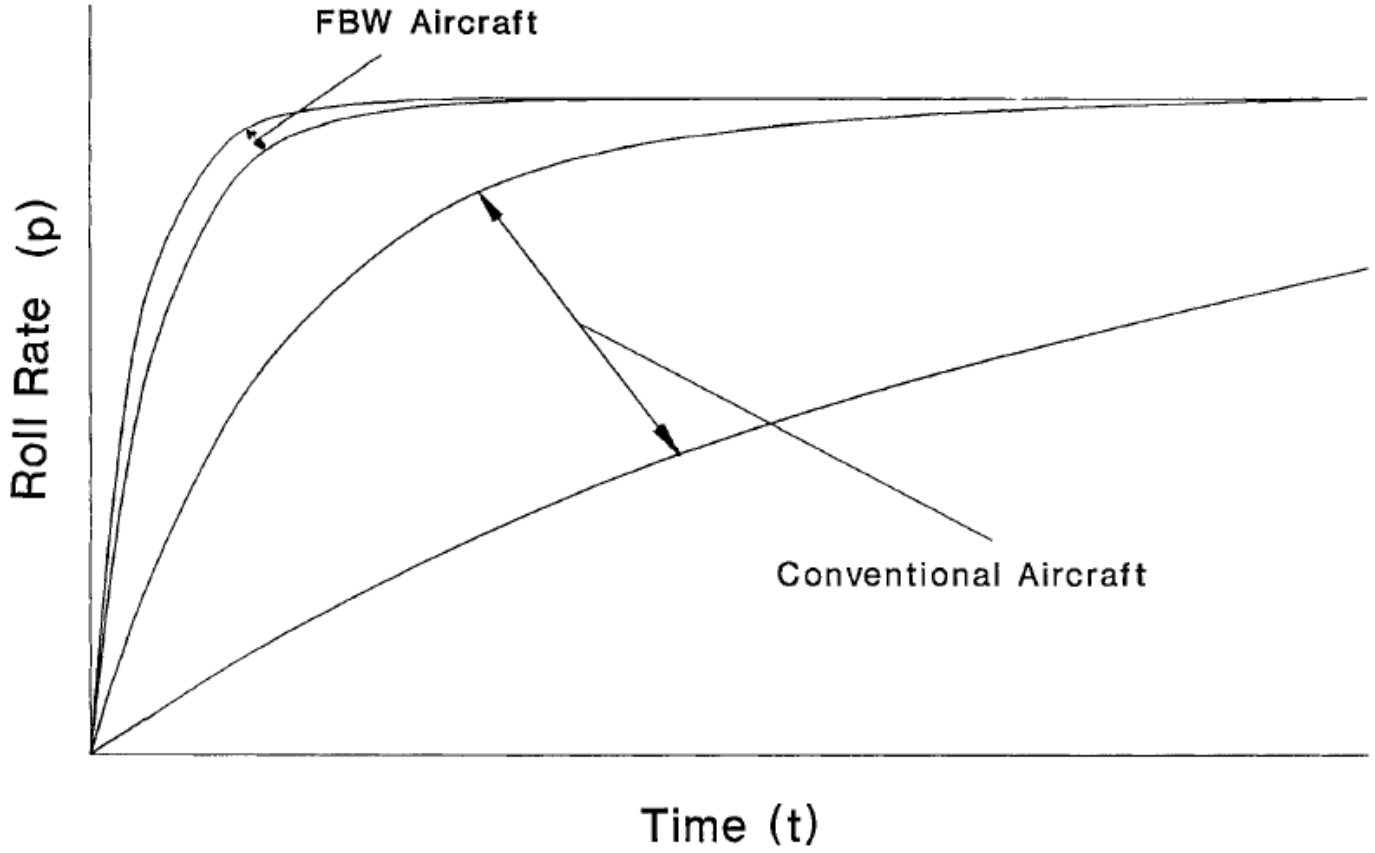


Fig. 4.8 Roll rate response.

Pitch Rate Command FBW Loop

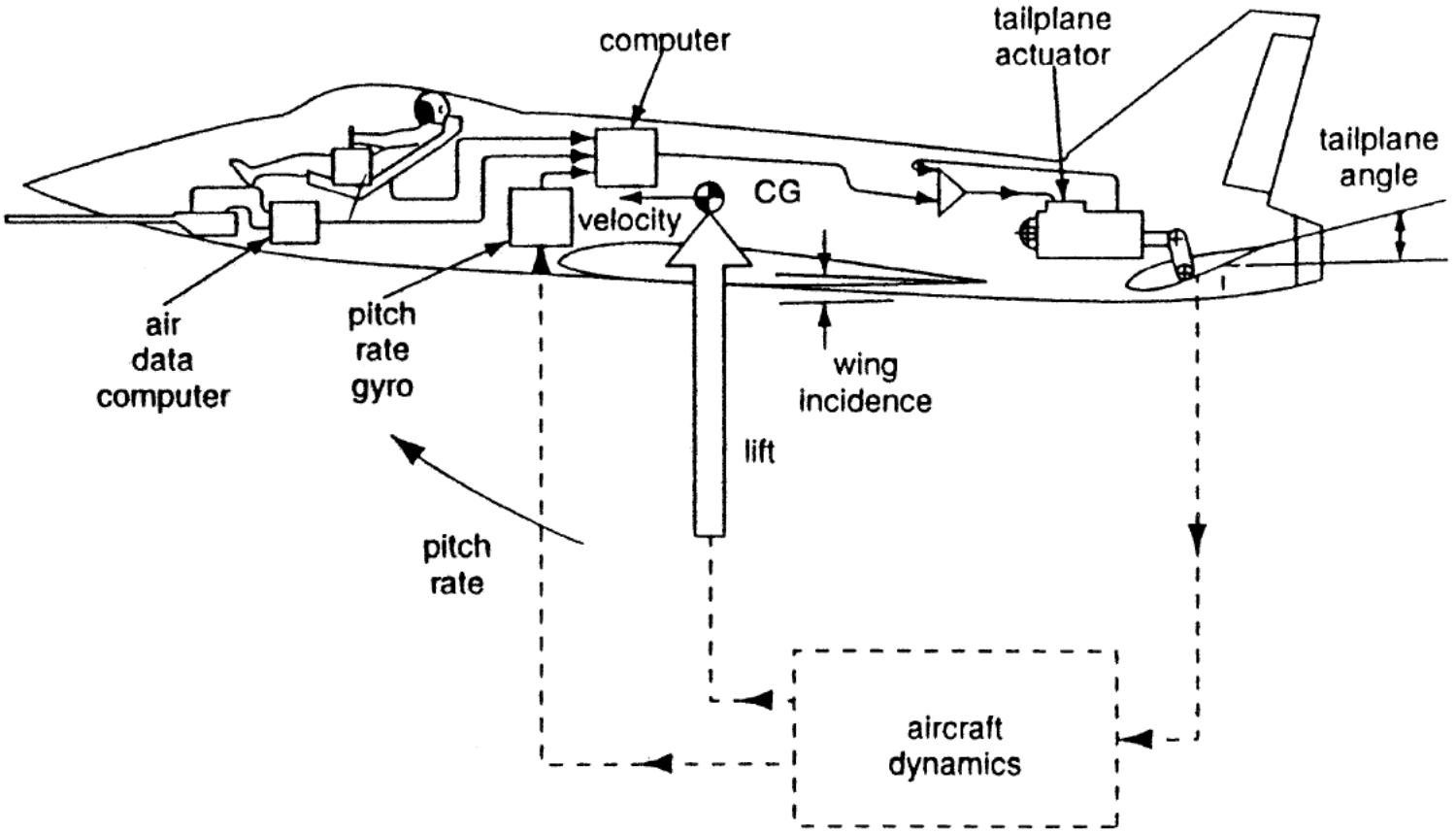


Fig. 4.9 Pitch rate command FBW loop.

Aerodynamics versus 'Stealth'

- Stealth considerations and requirements can conflict with aerodynamics requirements and FBW flight control is essential to give acceptable, safe handling across the flight envelope.

- Note 'flying wing' configuration and absence of vertical surfaces, yaw control being achieved by the operation of split ailerons.



Fig. 4.10 Northrop B2 'Spirit' strategic bomber.

Handing Quality

- The quality of the handling characteristics and the pilot workload are generally expressed in terms of the Cooper–Harper scale of ratings from one to ten (Table 1); one being excellent and ten the worst qualities possible.

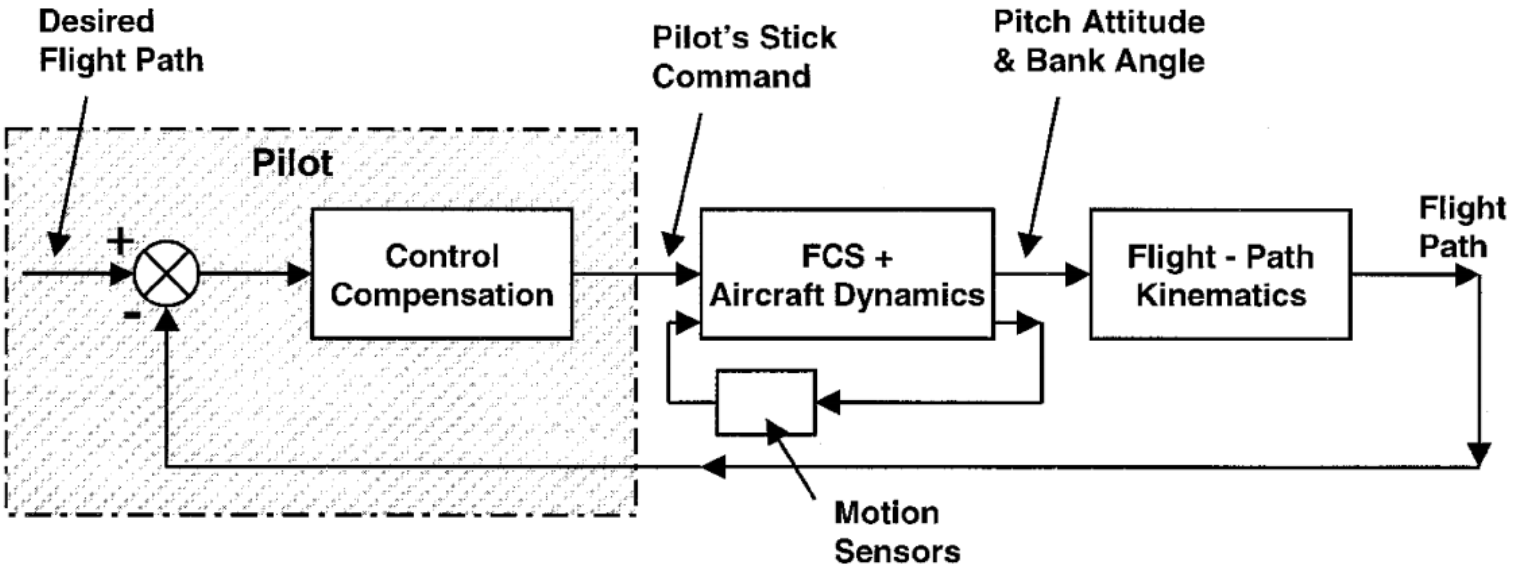
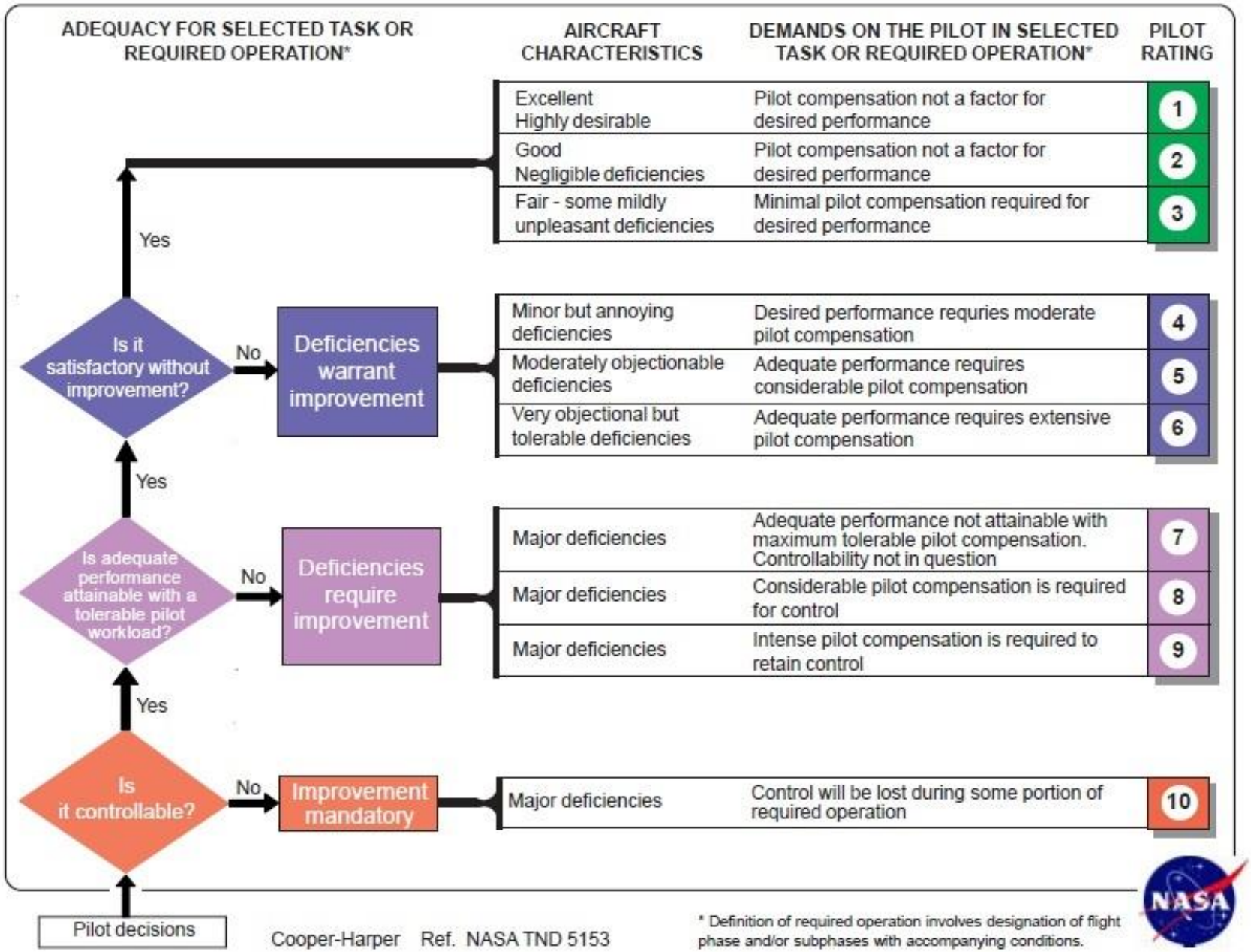


Fig. 4.25 Pilot in the loop.

COOPER-HARPER HANDLING QUALITIES RATING SCALE



Cooper-Harper Ref. NASA TND 5153

* Definition of required operation involves designation of flight phase and/or subphases with accompanying conditions.

Redundancy and Failure Survival

- Clearly the FBW flight control system must be no less safe than the simple mechanical control systems which it replaces.
- The safety levels required are specified in terms of the probability of a catastrophic failure occurring in the system from any cause whatsoever which could result in loss of control of the aircraft.
- It is generally specified that the probability of a catastrophic failure in the flight control system must not exceed
  - 1×10^{-7}/hour for a military aircraft
  - 1×10^{-9}/hour for a civil aircraft.

Redundant Configurations

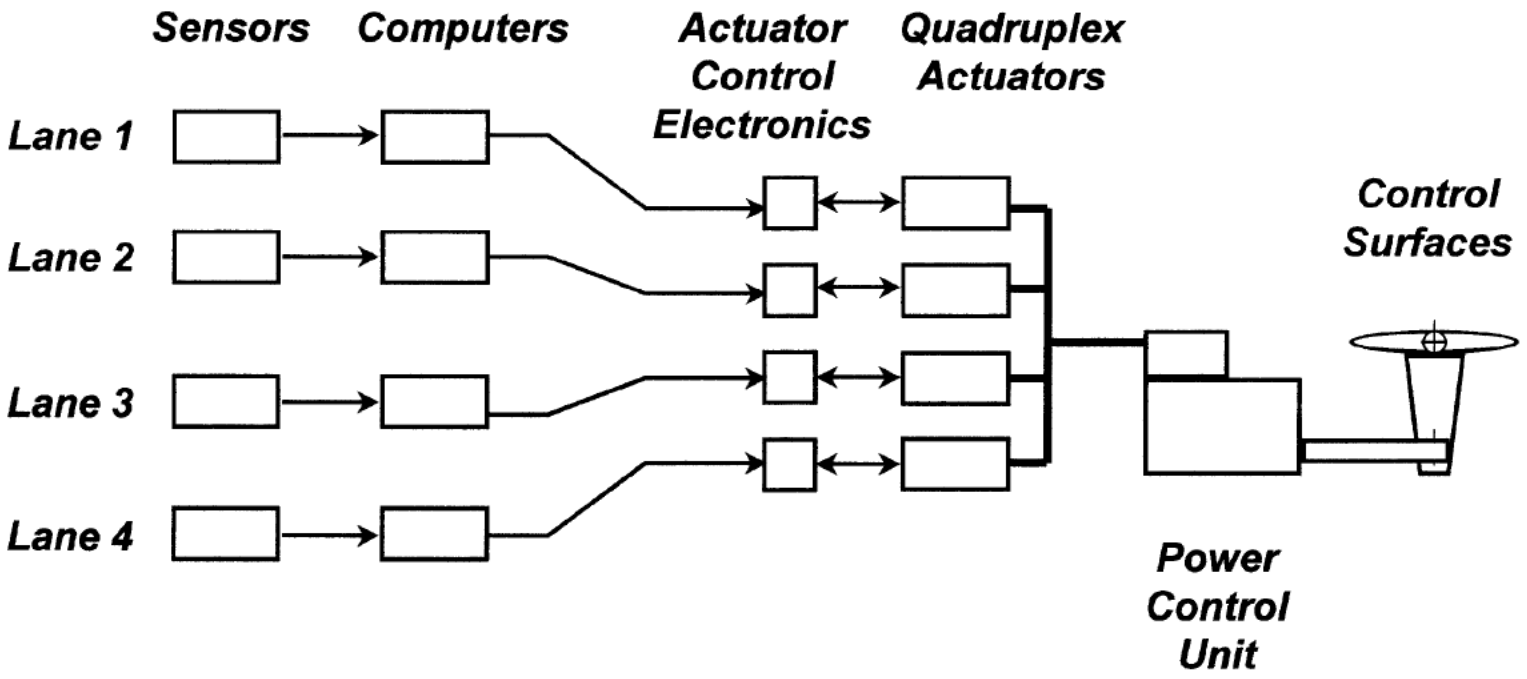


Fig. 4.26 Quadruplex system configuration. (Note: position feedback from control surface actuators omitted for clarity.)

Actuators in FBW

- To ensure stability and reliability for flight control, the actuators in the electronic terminals are multiple redundant design
- The operation computer determines which actuator to activate, carrying fault detection and recovery, also decide appropriate operating cycle and time for each actuator.

Quadruplex System Architecture

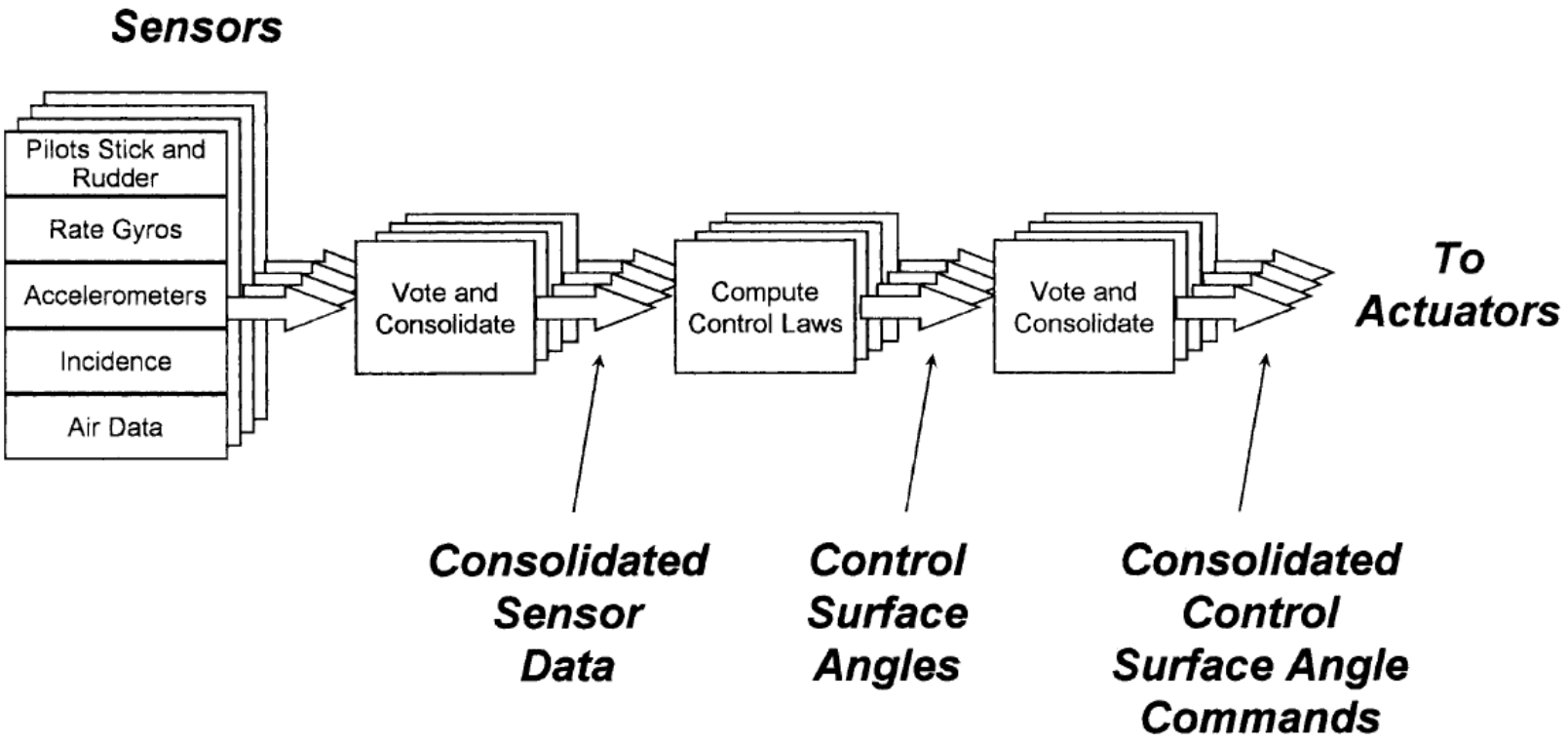


Fig. 4.31 Lane processing tasks.

Redundant Configurations

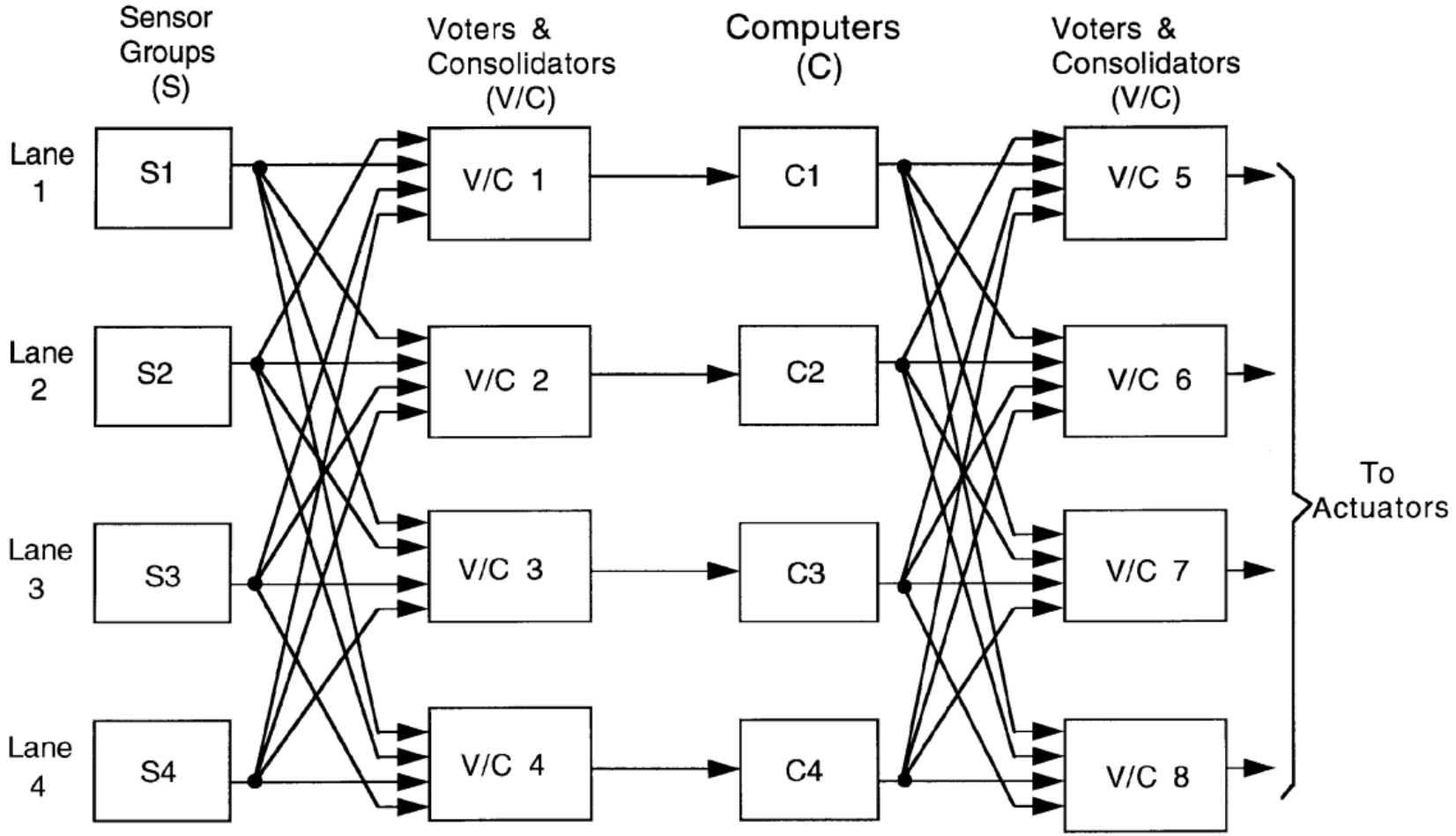


Fig. 4.32 Quadruplex system architecture.

A380 Flight Control Surfaces

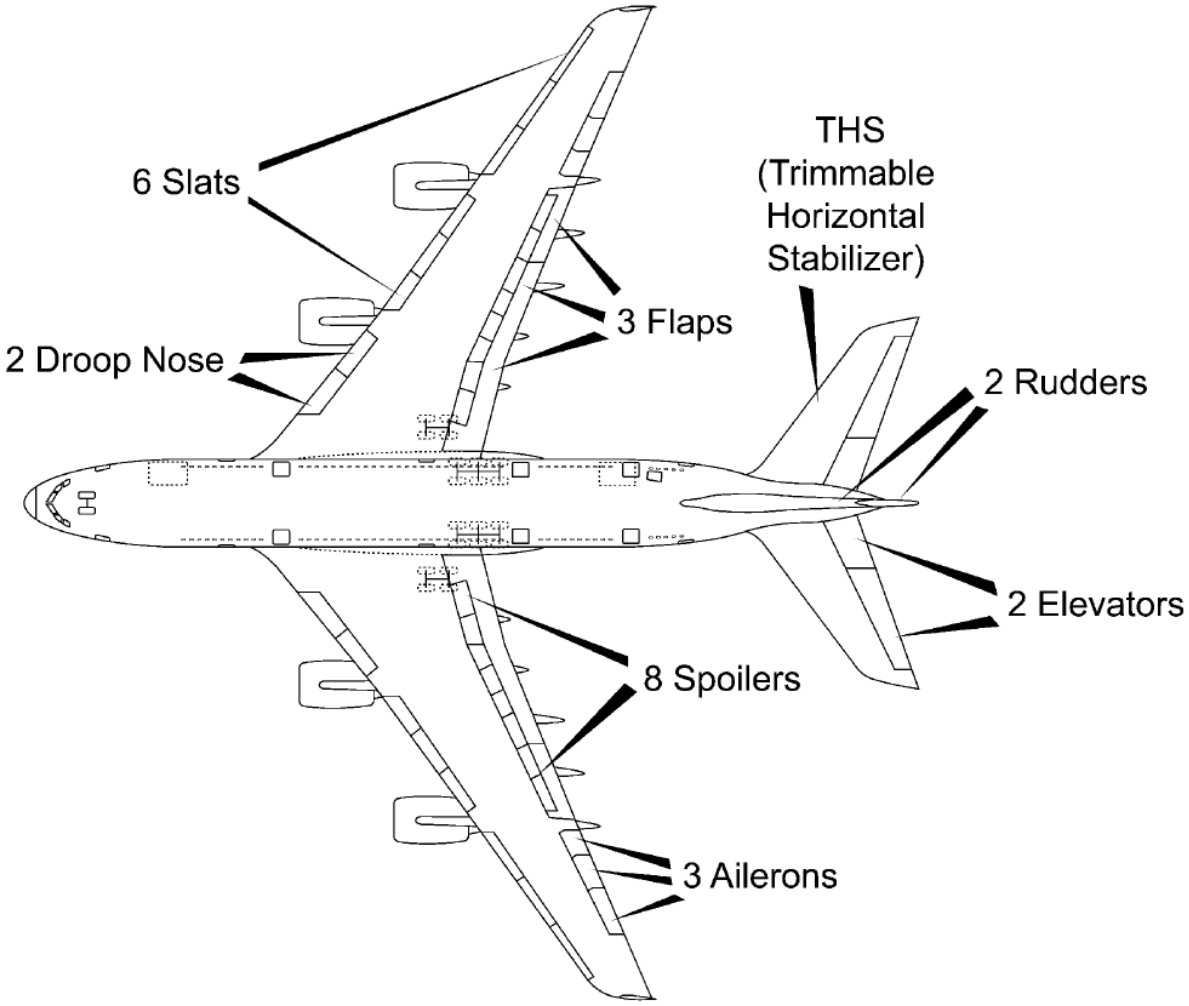


Fig. 4.34 A380 Flight control surfaces (by courtesy of Airbus).

A380 Flight Control Surfaces

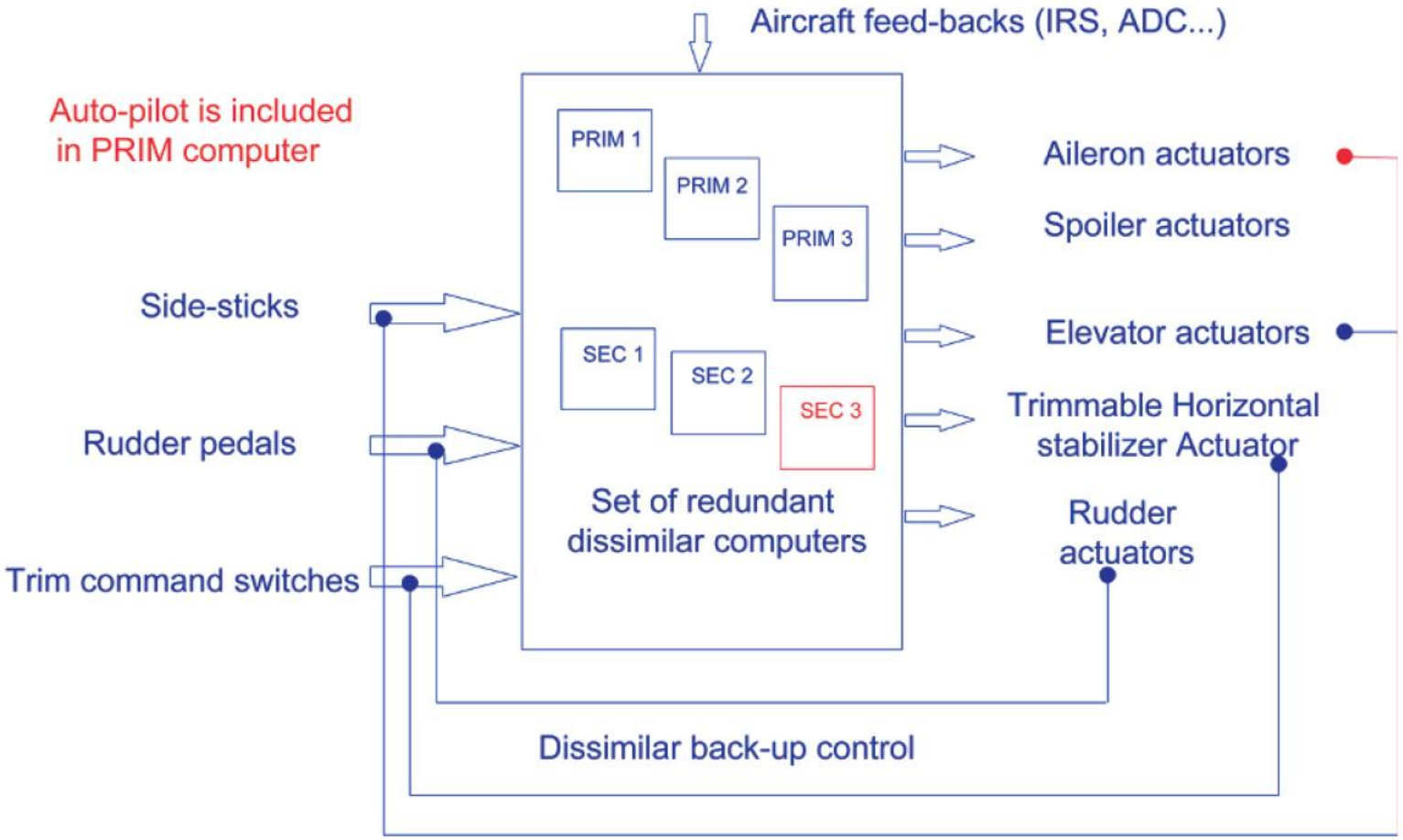


Fig. 4.35 A380 flight control system (by courtesy of Airbus).

Airbus A380 Flight Control System

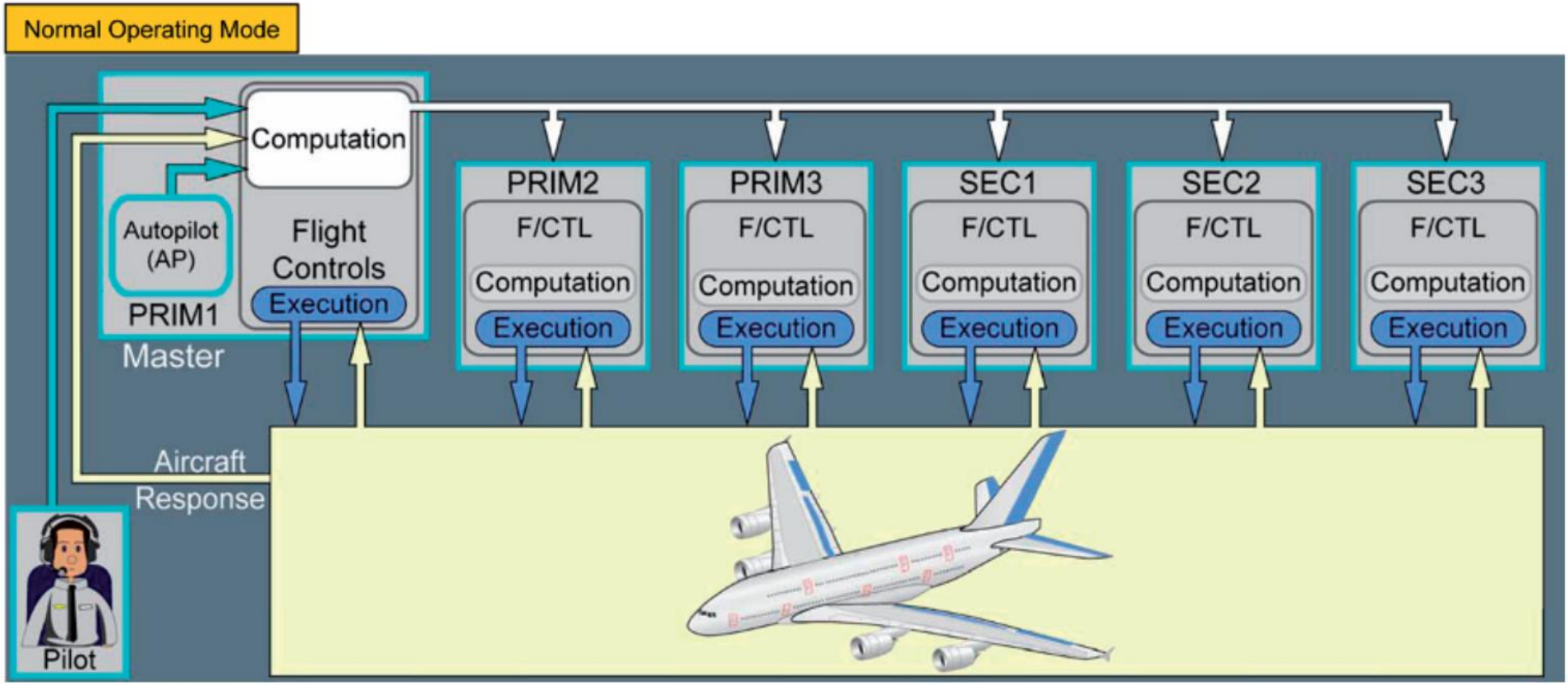


Fig. 4.36 Airbus A380 flight control system (by courtesy of Airbus).

Flight Control Software Functions

- The main functions that are carried out by the software for the flight control computers can be divided into three basic areas which have a degree of interaction and comprise:
  1. Redundancy management.
  2. Control laws.
  3. Built-in test.

Sensor Consolidation

- Consolidation of the data from a group of sensors to use a single representative value for subsequent computation of the control laws.
- A first-order list of sensors for a typical system is set out in Table 4.2 to show the numbers of sensors involved in these respective tasks. Most of these sensors would be at quadruplex or triplex level redundancy.

Table 4.2 Pitch channel sensors.

Pilot's control	Control stick	Pitch command sensors
		Roll command sensors
	Rudder bar sensors	
	Nose wheel steering sensors	
	Undercarriage position (discrete signal)	
Primary control surface positions	Tailplane angular position sensors	
	Elevator angular position sensors	
	Aileron angular position sensors	
	Rudder angular position sensors	
Secondary control surface positions	Wing spoiler angular position sensors	
	Leading edge flaps/slats angular position sensors	
	Trailingedge flaps angular position sensors	
Aircraft motion inertial sensors	Pitch rate gyros	
	Roll rate gyros	
	Yaw rate gyros	
	Normal accelerometers	
	Lateral accelerometers	
Air data sensors	Total pressure sensors	
	Static pressure sensors	
	Longitudinal airstream incidence sensors	
	Lateral airstream incidence sensors	